

### *Pretty Privilege: Is it Real?*

This project investigates whether “pretty privilege,” the idea that physically attractive individuals receive economic advantages, exists in the labor market. Specifically, we ask: Does physical attractiveness have a measurable effect on hourly wages? Using data from the Wooldridge repository, which includes subjective ratings of physical attractiveness on a five-point scale, we analyze whether higher attractiveness scores are associated with higher hourly earnings. This question is important for understanding potential biases in wage determination and the implications for meritocracy in hiring and compensation practices.

Our data comes from the Wooldridge repository’s (n=1,260) cross-sectional labor market dataset, which contains individual-level information on hourly wages and physical attractiveness, as rated by independent observers. Attractiveness is measured on a scale from 1 (least attractive) to 5 (most attractive). The dataset also includes standard controls such as education, experience, and demographic characteristics, allowing us to isolate the effect of attractiveness on wages while accounting for other relevant factors.

To test our hypothesis that higher physical attractiveness leads to higher wages, we estimate a linear regression of log hourly wage on the attractiveness score, controlling for education, experience, and good health ( $\ln(\text{Wage}) = \alpha + \beta_{\text{Looks}} + \beta_{\text{Exper}} + \beta + \text{Educ} + \beta_{\text{GoodHlth}} + \epsilon$ ). Our analysis finds a statistically significant positive relationship: for each one-point increase in attractiveness, hourly wages rise by approximately 5.7%. These findings support the existence of pretty privilege and suggest that physical appearance plays a meaningful role in wage determination, independent of other qualifications. However, it is important to note the existence of omitted variable bias: Traits like confidence, grooming, and social skills may enhance perceived attractiveness and independently affect earnings. If these traits are unobserved in our data, they could cause:

- Upward bias: If attractive people disproportionately possess these traits (e.g., confidence), our estimate overstates the pure “beauty” effect.
- Downward bias: If attractiveness compensates for lacking these traits (e.g., introversion), our estimate understates the true effect.

Our results highlight a persistent, measurable effect of physical attractiveness on economic outcomes, raising important questions about fairness and bias in the labor market. Further research could explore the mechanisms behind this effect and potential policy responses. Also, the legitimacy of our model with unexplored omitted variable bias.

```
Call:
lm(formula = lwage ~ looks + goodhlth + educ + exper, data = beauty)

Residuals:
    Min       1Q   Median       3Q      Max
-1.7033 -0.3152  0.0052  0.3317  2.7833

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.097553   0.115938   0.841   0.4003
looks        0.056920   0.022251   2.558   0.0106 *
goodhlth     0.125364   0.060570   2.070   0.0387 *
educ         0.072900   0.005855  12.451  <2e-16 ***
exper        0.019061   0.001288  14.795  <2e-16 ***
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5295 on 1255 degrees of freedom
Multiple R-squared:  0.2092,    Adjusted R-squared:  0.2067
F-statistic:    83 on 4 and 1255 DF,  p-value: < 2.2e-16
```

See Table 1 for a summary of regression results and Appendix A for additional robustness checks.

Table 1. Estimated Effect of Attractiveness on Hourly Wage

| Attractiveness Score<br>Increase | Marginal Increase in Hourly<br>Wage | Annualized Wage Increase<br>(Full-Time) |
|----------------------------------|-------------------------------------|-----------------------------------------|
| +1                               | \$2.06                              | \$4,284.80                              |

(Regression controls for education, experience, and demographics. See Appendix A for details.)

#### Appendix A: Robustness Check, OVB, and Subjectivity

##### 1. Age as a Confounder

Age correlates with both declining attractiveness and wage changes. Controlling for age, our wage regression shows a stable attractiveness effect (0.057,  $p < 0.001$ ), ruling out age-driven bias.

##### 2. Subjectivity of Beauty Ratings

Using averaged ratings from multiple raters or simulated scores, the attractiveness coefficient remains significant (0.066 – 0.076,  $p < 0.001$ ), confirming robustness to rater bias.

Conclusion: While beauty predicts wages in our model, observational data cannot isolate causality. A more causal design (e.g., natural experiments with random resume photos) would be needed to rule out OVB.